

S $\Delta\phi$ -24 — Authority and Innovation as Dual Regimes: Minimal Axioms for Compression, Re-opening, and Update Triggers in Coupled Environments (v1.0)

Sofience — Working paper — 2026

Core thesis: Authority is compressed low-resistance convergence; innovation is misfit-triggered re-opening; conflict is a phase signal, not a moral dispute.

This paper formalizes authority and innovation as dual regimes within the S $\Delta\phi$ framework. Authority emerges as compression-driven convergence toward low-resistance routes ($\Delta\phi\downarrow$), while innovation emerges as misfit-triggered re-opening ($\Delta\phi\uparrow$) when accumulated mismatch forces routes to become computable again. The central claim is that authority-innovation conflict is not a moral disagreement but a phase transition signal. In co-environmentalization (S $\Delta\phi$ -23), this conflict propagates across coupled systems and becomes a governance problem.

Keywords: authority, innovation, compression, re-opening, misfit, triggers, authority debt, closure, drift, coupled environments, S $\Delta\phi$.

1. Problem

SΔφ-21 defined judgment as authority assignment. SΔφ-22 defined authority editing (AEP). SΔφ-23 extended authority into coupled environments and installed ontology-axiology separation as a safety boundary. The next question is structural: why does authority inevitably form, and why does innovation appear as conflict? Further, when does conflict become re-opening, and when is it sealed as heresy?

2. Minimal Definitions

Term	Definition
Auth (Authority)	A low-resistance default route (attractor) produced by accumulated compression
C (Compression)	Route-space reduction that lowers $\Delta\phi$ variability and execution cost ($\Delta\phi\downarrow$).
Inv (Innovation)	A re-opening event where alternatives become computable again ($\Delta\phi\uparrow$).
O (Re-opening)	Transition that restores computability of previously pruned routes.
ϵ (Misfit)	Mismatch signal between model and environment; accumulated coherence failure
τ (Trigger)	Threshold event activating re-opening: conflict, error, non-resonance, or external
DA (Authority debt)	Revision resistance accumulated by rigid authority; increases edit cost over time
CE amplification	In coupled environments, ϵ , τ , and DA propagate and amplify across systems.

3. Minimal Axioms (Authority-Innovation Duality)

AI1 (Compression produces authority). Accumulated compression yields a default attractor route (authority).

AI2 (Authority reduces $\Delta\phi$). Authority lowers $\Delta\phi$ variability and reduces decision/transition cost.

AI3 (Over-compression accumulates misfit). Excessive compression accumulates mismatch ϵ under environmental change or exception buildup.

AI4 (Innovation is misfit-triggered re-opening). When ϵ crosses a threshold, re-opening O is forced, appearing as innovation.

AI5 (Conflict is phase signal). Authority-innovation conflict is a transition signal τ , not a moral dispute.

AI6 (Coupled propagation in CE). In CE, rigidity or re-opening on one side shifts ϵ , τ , DA on the other side and can trigger system-level regime shifts.

4. Mechanism: Compression Loop, Misfit Loop, Re-opening Loop

Authority and innovation alternate as phases of the same dynamics:

Loop	Sketch	Outcome
Compression loop	repeat selection $\rightarrow$ prune alternatives $\rightarrow$ cost $\downarrow$	Auth forms; $\Delta\phi\downarrow$
Misfit loop	Auth rigidifies $\rightarrow$ exceptions accumulate $\rightarrow$ $\epsilon\uparrow$	τ approaches; DA $\uparrow$

Re-opening loop	τ triggers AEP $\rightarrow$ alternatives computable $\rightarrow \Delta\phi \uparrow$	Inv appears; new route search
Outcome split	(a) re-compress successfully / (b) seal & suppress	new authority / closure precursor

5. Safety Note: Ontology-Axiology Separation under Authority

Authority often amplifies irreversible trace strength (S_e), especially in coupled environments. However $S\Delta\phi$ -23 enforces a safety boundary: trace strength is not value. If strong authority is treated as moral legitimacy, innovation is automatically sealed as heresy and closure becomes justified.

Rule: $S_e \neq V_o$. Conflict τ must be treated as potential editing material ($S\Delta\phi$ -22), not as moral failure by default.

6. Diagnostic Regimes

Stable authority. ε low; $\Delta\phi \downarrow$; editability E^d preserved.

Rigid authority (closure precursor). $\varepsilon \uparrow$ but τ suppressed/reframed; $DA \uparrow$; re-opening becomes expensive or impossible.

Innovative re-opening. τ accepted; $\Delta\phi \uparrow$; alternatives computed; short-term instability tolerated.

Suppressed innovation. innovation attempts are sealed as illegitimate; misfit accumulates; long-term collapse risk increases.

Coupled regime shift (CE). one side's rigidity or re-opening propagates and shifts the coupled system's phase.

7. Series Position and References

$S\Delta\phi$ -24 formalizes the authority-innovation duality as a dynamics of compression and re-opening, and uses the protocol layer ($S\Delta\phi$ -22) and coupling layer ($S\Delta\phi$ -23) to distinguish re-opening from suppression.

Series references (DOI):

$S\Delta\phi$ -23 — Co-environmentalization Dynamics: Coupled Authority, Reality Constraints, and the Ontology-Axiology Split (v1.0) — <https://doi.org/10.5281/zenodo.18801221>

$S\Delta\phi$ -22 — Authority Editing Protocols: Minimal Re-entry Governance after Judgment (v1.0) — <https://doi.org/10.5281/zenodo.18800024>

$S\Delta\phi$ -21 — Judgment as Intuition Authority Assignment (v1.0) — <https://doi.org/10.5281/zenodo.18798295>

$S\Delta\phi$ -20 — Intuition as Irreversible Compression Trace (v1.0) — <https://doi.org/10.5281/zenodo.18787647>

$S\Delta\phi$ -17 — Openness Stabilization: Path Principles... (v1.0) — <https://doi.org/10.5281/zenodo.18763581>

$S\Delta\phi$ -16 — Evil as Relational Closure (v1.0) — <https://doi.org/10.5281/zenodo.18763262>

8. Exit Questions

EQ1. When does conflict τ become re-opening O, and when is it sealed as heresy?

EQ2. At what threshold of DA does authority rigidity become practically irreversible?

EQ3. In CE, what minimal conditions allow innovation to open the coupled system rather than collapse it into closure?